

Transaction Rule Engine — API Reference

Spring Boot + Drools (DRL) · runtime rule & flow management · Base URL: `http://localhost:8080`

Concept: Rules (DRL) and routing (flows + context mappings) live in the database and are managed entirely through these APIs — no restart. A request builds a `ValidationContext`, its class is looked up to find the flow, the flow lists ordered agenda groups, and the active rules in those groups run.

1. Quick setup (fresh deploy)

Run these three once after startup, then the transaction API works. Order matters.

STEP 1 — UPLOAD RULES

```
curl.exe -X POST http://localhost:8080/admin/rules/upload -F "file=@transaction-rules.drl"
```

STEP 2 — DEFINE THE FLOW (FLOW → ORDERED GROUPS)

```
curl.exe -X POST http://localhost:8080/admin/flows -H "Content-Type: application/json" -d "{\"flowName\":\"TRANSFER_TRANSACTION\",\"groups\":[{\"COMMON\"},{\"TRANSFER\"}]}"
```

STEP 3 — MAP THE CONTEXT CLASS → FLOW

```
curl.exe -X POST http://localhost:8080/admin/flows/mappings -H "Content-Type: application/json" -d "{\"contextClass\":\"com.example.droolspec.model.ValidationContext\",\"flowName\":\"TRANSFER_TRANSACTION\"}"
```

2. Transaction API

POST /api/transactions/validate

Validate a transaction against the active rules.

HEADERS

Content-Type: application/json

REQUEST BODY

```
{
  "userId": "USER-001",
  "transactionMode": "TRANSFER",
  "debitCredit": "DR",
  "sourceAccount": "100001",
  "currency": "BDT",
  "amount": 500
}
```

REQUEST FIELDS

Field	Type	Required	Notes
userId	string	yes	lookup key for user limit + blocked check
transactionMode	string	yes	TRANSFER CREDIT
debitCredit	string	yes	DR CR
sourceAccount	string	yes	lookup key for product
currency	string	yes	e.g. BDT
amount	number	yes	must be > 0
sourceBranch, destinationAccount, exchangeRate, remarks	-	no	optional / informational

RESPONSE 200 — BLOCKED (OVER LIMIT)

```
{
  "valid": false,
  "permissionDenied": true,
  "message": "Blocked: amount exceeds limit on a debit-restricted BDT account"
}
```

RESPONSE 200 — ALLOWED

```
{
  "valid": true,
  "permissionDenied": false,
  "message": null
}
```

RESPONSE 400 — INVALID INPUT

```
{
  "timestamp": "2026-09-29T10:00:00Z",
  "status": 400,
  "error": "Validation failed",
  "fieldErrors": { "amount": "amount must be greater than 0" }
}
```

RESPONSE 422 — MISSING CONTEXT (NO USER LIMIT / PRODUCT ROW)

```
{
  "timestamp": "2026-09-29T10:00:00Z",
  "status": 422,
  "error": "Context not found",
  "message": "No product for sourceAccount=999999"
}
```

CURL

```
curl.exe -X POST http://localhost:8080/api/transactions/validate -H "Content-Type: application/json" -d "{\"userId\":\"USER-001\",\"transactionMode\":\"TRANSFER\", \"debitCredit\":\"DR\", \"sourceAccount\":\"100001\", \"currency\":\"BDT\", \"amount\":500}"
```

3. Rules API

/admin/rules — manage DRL files at runtime. Every change validates (compiles) first; a bad DRL is rejected and the running rules are unchanged.

GET /admin/rules

List all rule files.

RESPONSE 200

```
[
  { "fileName": "transaction-rules.drl", "active": true, "version": 1, "updatedAt": "2026-09-29T10:00:00Z" }
]
```

CURL

```
curl.exe http://localhost:8080/admin/rules
```

POST /admin/rules/upload

Upload/replace a DRL by file (multipart). Auto-reloads. Best for real .drl files.

FORM FIELD

file = <the .drl file> (multipart/form-data)

RESPONSE 200

```
{ "status": "reloaded", "fileName": "transaction-rules.drl" }
```

CURL

```
curl.exe -X POST http://localhost:8080/admin/rules/upload -F "file=@transaction-rules.drl"
```

POST /admin/rules

Upload/replace a DRL by JSON. Auto-reloads.

REQUEST BODY

```
{
  "fileName": "extra-rules.drl",
  "drl": "package com.example.droolspec.rules\nrule \"x\" when then end\n"
}
```

RESPONSE 200

```
{ "status": "reloaded", "fileName": "extra-rules.drl" }
```

RESPONSE 400 — BAD DRL (NOTHING SAVED, OLD RULES KEPT)

```
{
  "timestamp": "2026-09-29T10:00:00Z",
  "status": 400,
  "error": "Rule compilation failed",
  "message": "DRL compilation failed: [Message [id=1, ...]]"
}
```

CURL

```
curl.exe -X POST http://localhost:8080/admin/rules -H "Content-Type: application/json" -d "{\"fileName\":\"extra-rules.drl\",\"drl\":\"package com.example.drool\nspec.rules\nrule \\\"x\\\" when then end\n\"}"
```

DELETE /admin/rules/{fileName}

Deactivate a rule file. Auto-reloads.

RESPONSE 200

```
{ "status": "reloaded", "fileName": "extra-rules.drl" }
```

CURL

```
curl.exe -X DELETE http://localhost:8080/admin/rules/extra-rules.drl
```

4. Flows API

/admin/flows — a flow is a name plus an ordered list of agenda groups. Every change auto-reloads.

GET /admin/flows

List flows and their ordered groups.

RESPONSE 200

```
{ "TRANSFER_TRANSACTION": ["COMMON", "TRANSFER"] }
```

CURL

```
curl.exe http://localhost:8080/admin/flows
```

POST /admin/flows

Create or replace a flow's ordered groups.

REQUEST BODY

```
{
  "flowName": "TRANSFER_TRANSACTION",
  "groups": ["COMMON", "TRANSFER"]
}
```

RESPONSE 200

```
{ "status": "reloaded", "flow": "TRANSFER_TRANSACTION", "groups": ["COMMON","TRANSFER"] }
```

CURL

```
curl.exe -X POST http://localhost:8080/admin/flows -H "Content-Type: application/json" -d "{\"flowName\":\"TRANSFER_TRANSACTION\",\"groups\":[\"COMMON\",\"TRANSFER\"]}"
```

DELETE /admin/flows/{flowName}

Delete a flow (removes its groups). Auto-reloads.

RESPONSE 200

```
{ "status": "reloaded", "flow": "TRANSFER_TRANSACTION" }
```

CURL

```
curl.exe -X DELETE http://localhost:8080/admin/flows/TRANSFER_TRANSACTION
```

GET /admin/flows/overview

The whole routing picture: mappings + flows together.

RESPONSE 200

```
{
  "mappings": [
    { "contextClass": "com.example.droolspec.model.ValidationContext", "flowName": "TRANSFER_TRANSACTION" },
  ],
  "flows": { "TRANSFER_TRANSACTION": ["COMMON", "TRANSFER"] }
}
```

CURL

```
curl.exe http://localhost:8080/admin/flows/overview
```

POST /admin/flows/reload

Manually reload flow config into memory (writes above already auto-reload).

RESPONSE 200

```
{ "status": "flow config reloaded" }
```

CURL

```
curl.exe -X POST http://localhost:8080/admin/flows/reload
```

5. Context mappings API

/admin/flows/mappings — map a context (fact) class to a flow. One class → one flow.

GET /admin/flows/mappings

RESPONSE 200

```
[
  { "contextClass": "com.example.droolspec.model.ValidationContext", "flowName": "TRANSFER_TRANSACTION" }
]
```

CURL

```
curl.exe http://localhost:8080/admin/flows/mappings
```

POST /admin/flows/mappings

Create or re-map a context class to a flow. Auto-reloads.

REQUEST BODY

```
{
  "contextClass": "com.example.droolspec.model.ValidationContext",
  "flowName": "TRANSFER_TRANSACTION"
}
```

RESPONSE 200

```
{ "status": "reloaded", "contextClass": "com.example.droolspec.model.ValidationContext", "flow": "TRANSFER_TRANSACTION" }
```

CURL

```
curl.exe -X POST http://localhost:8080/admin/flows/mappings -H "Content-Type: application/json" -d '{"contextClass":"com.example.droolspec.model.ValidationContext","flowName":"TRANSFER_TRANSACTION"}'
```

DELETE /admin/flows/mappings?contextClass={class}

Remove a mapping. Auto-reloads.

RESPONSE 200

```
{ "status": "reloaded", "contextClass": "com.example.droolspec.model.ValidationContext" }
```

CURL

```
curl.exe -X DELETE "http://localhost:8080/admin/flows/mappings?contextClass=com.example.droolspec.model.ValidationContext"
```

6. Health

GET /actuator/health

RESPONSE 200

```
{ "status": "UP" }
```

CURL

```
curl.exe http://localhost:8080/actuator/health
```

Endpoint summary

Method	URL	Purpose
POST	/api/transactions/validate	Validate a transaction
GET	/admin/rules	List rule files
POST	/admin/rules/upload	Upload DRL (multipart)
POST	/admin/rules	Upload DRL (JSON)
DELETE	/admin/rules/{fileName}	Deactivate rule file
GET	/admin/flows	List flows → groups
POST	/admin/flows	Create/replace flow

DELETE	/admin/flows/{flowName}	Delete flow
GET	/admin/flows/overview	Whole routing picture
POST	/admin/flows/reload	Reload flow config
GET	/admin/flows/mappings	List context mappings
POST	/admin/flows/mappings	Map context class → flow
DELETE	/admin/flows/mappings	Remove mapping
GET	/actuator/health	Health check